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Tutorial 8

2. Exercise 1

Document1: D1: 'The cat chased the mouse".

Document 2: D2: "The dog barked at the cat."

Documents - D3 : "The mouse ran away from the cat".

Step 1: Tokenization and preprocessing.

1. Convert all text to lowercase.

D1: " the cat chased the mouse".

D2: " the dog barked at the cat."

D3: " the mouse ran away from the cat."

2. Remove punctuations - X.

3. Tokenize each document into individual words (token).

D1 : "the cat chased the mouse"

D2 : "the dog barked at the cat."

D3 : "the mouse ran away from the cat"

4. Remove stop words.

D1: "cat chased mouse"

D2: "dog barked cat"

D3: "mouse ran cat"

STEP 2: vocabulary construction.

List of tokens: ["barked", "cat", "chased", "dog", "mouse", "ran"]

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Step 3:- Term frequency calculations.

vocab	D1	D2	D3.
barked	0/3 = 0	1/3 = 0.33	0/3. 0.
cat	1/3 = 0.33	1/3 = 0.33	1/3 = 0.33
chased	1/3 = 0.33	0/3 = 0	0/3 = 0
dog	0/3 = 0	1/3 = 0.33	0/3 = 0.
mouse	1/3 = 0.33	0/3 = 0	1/3 = 0.33
ran	0/3 = 0	0/3 = 0	1/3 = 0.33.

Step 3:- 1 Normalized TF $w_{TF}(t,d) = \begin{cases} 1+\log(tf_{t,d}) & \text{if } tf_{t,d} > 0 \\ 0 & \text{otherwise} \end{cases}$

vocab	D_1	D_2	D_3
barked	0	0.52	0
cat	0.52	0.52	0.52
chased	0.52	0	0
dog	0	0.52	0
mouse	0.52	0	0.52
ran	0	0	0.52

Step4: Inverse document frequency (IDF) calculation.

$IDF(t) = \log\left(\frac{N}{1+df_t}\right)$ N: no. of document

df_t: no. of doc where t appears

vocab	IDF
barked	$\log\left(\frac{3}{1+1}\right) = 0.176$
cat	$\log\left(\frac{3}{1+3}\right) = 0.124$
chased	$\log\left(\frac{3}{1+1}\right) = 0.176.$
dog	$\log\left(\frac{3}{1+1}\right) = 0.176.$
mouse	$\log\left(\frac{3}{1+2}\right) = 0$
ran	$\log\left(\frac{3}{1+1}\right) = 0.176.$

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Step 5 :- Compute TF - IDF vectors.

$$TF\text{-}IDF(t,d) = TF_{t,d} \times (IDF(t))$$

Vocab	D_1	D_2	D_3
barked	0	0.091	0
cat	-0.064	-0.064	-0.064
chased	0.091	0	0
dog	0	0.091	0
mouse	0	0	0
ran.	0	0	0.091

3. "word 2vec"

3.1 Exercise 2 : Data Generation from CBOW.

D1 : "The cat chased the mouse"

D2 : "The dog Barked at the cat".

D3 : "The mouse ran away from the cat".

After stop-words removal.

D1 : ["cat", "chased", "mouse"]

D2 : ["dog", "barked", "cat"]

D3 : ["mouse", "ran", "cat"]

vocabulary:

["barked", "cat", "chased", "dog", "mouse", "ran"]

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Indexing.

Word	Index	One-hot vector.
barked	0	[1, 0, 0, 0, 0, 0]
cat	1	[0, 1, 0, 0, 0, 0]
chased	2	[0, 0, 1, 0, 0, 0]
dog	3	[0, 0, 0, 1, 0, 0]
mouse	4	[0, 0, 0, 0, 1, 0]
ran	5	[0, 0, 0, 0, 0, 1]

Step 2: Look up table.

for D1: ["cat", "chased", "mouse"]

context (words)	context (index)	Target word	Target (Index)
chased	[2]	cat	1
cat, mouse	[1, 4]	chased	2
chased	[2]	mouse	4

for D2: ["dog", "barked", "cat"]

context (words)	context (index)	Target word	Target (Index)
barked	[0]	dog	3
dog, cat	[3, 1]	barked	0
barked	[0]	cat	1

for D3: ["mouse", "ran", "cat"]

context	context (index)	Target (word)	Target (index)
ran	[5]	mouse	4
mouse, cat	[4, 1]	ran	5
ran	[5]	cat	1